

EOS Pub Operating and Assurance System

Antigravity Master Build Specification

Version 3.0

Date 13 September 2026

Status Build ready specification

Primary deployment United Kingdom with an England ruleset first

Primary client Mobile first installable progressive web application

Source basis EOS SPEC Master System Input V2 plus the expanded pub operating system requirements

1 Build instruction

Build the production system described in this document. Treat every statement containing **MUST** as an acceptance requirement, **SHOULD** as a strong default, **MAY** as optional, and **FUTURE** as outside the first production release.

Do not replace concrete requirements with generic placeholders. Do not mark a feature complete because a screen exists. A feature is complete only when its workflow, validation, permissions, offline behaviour, audit evidence, error handling, tests, and accessible user interface work together.

Work in the delivery phases defined in this specification. At the end of each phase:

1. run unit, integration, end to end, permission, accessibility, offline, and sync tests that apply to that phase;
2. repair all failing tests and type errors;
3. update the database migrations, seed data, API documentation, decision records, and requirement traceability;
4. produce a working preview using realistic demonstration data;
5. record unresolved decisions without silently inventing legal, commercial, or premises-specific facts.

The application must remain usable if the internet disappears during a shift. An enrolled user must be able to unlock an approved device, complete checklists, record readings, capture photographs, log incidents, receive cash orders, count stock, and continue operational work. New records and attachments must enter a durable encrypted queue and upload automatically when an internet connection becomes available.

2 Product definition

EOS Pub is a multi-tenant operating and assurance system for pubs. It combines the live control of a shift with stock, purchasing, recipes, POS, finance, food safety, licensing, people, training, assets, maintenance, utilities, carbon, waste, incidents, events, guest feedback, executable procedures, evidence, and organisational knowledge.

The product must let a suitably authorised and competent person enter a premises, take control of a shift, and see:

- what must be done now;
- how to do it;
- which tasks are blocked and why;
- what a passing result looks like;
- which evidence is required;
- which corrective action applies to a failure;
- who must be told;
- whether the premises is ready to open, safe to continue trading, or ready to close;
- where money, stock, energy, and time are being lost.

EOS Pub does not make an untrained person competent. It must prevent users from carrying out controlled tasks when their role, training, local authorisation, or current competency does not permit the task.

3 Product outcomes

The production system **MUST**:

- create one operational record from each sale, delivery, stock movement, waste event, checklist response, reading, approval, incident, maintenance action, and shift event;
- maintain theoretical stock and compare it with physical stock in units, volume, cost, retail value, and variance percentage;
- maintain price history and recalculate product cost and margin after supplier price changes;
- calculate daily and weekly profit indicators using sales, cost of goods, waste, stock variance, discounts, labour, and energy;
- measure electricity, gas, water, cost, and kgCO₂e with versioned conversion factors;
- turn applicable legal, licence, safety, and company rules into scheduled or triggered tasks with evidence and escalation;
- provide an executable checklist and sign-off for every operational procedure;
- support first-run onboarding detailed enough to configure a real premises;
- provide secure sign-up, login, invitation, role assignment, MFA, device enrolment, offline unlock, and account recovery;

- store operational work locally when offline and synchronise it safely when online;
- show sync status and never imply that unsynchronised data is safely held by the server;
- preserve an immutable audit history for controlled and financial events;
- export evidence packs suitable for review by managers, accountants, insurers, auditors, and inspecting authorities;
- meet WCAG 2.2 AA for the application workflows in scope.

4 Scope and release priorities

4.1 Production release scope

P0 is required for the first production release:

- organisation and premises onboarding;
- authentication, MFA, user invitations, roles, device management, and offline unlock;
- offline application shell, local database, evidence capture, outbox, sync, conflict handling, and recovery;
- Run the Pub shift control board;
- universal rules, tasks, executable SOPs, evidence, exceptions, corrective actions, escalations, and sign-offs;
- premises compliance profile and England compliance source register;
- competency passports and task unlocking;
- opening, service, handover, and closing workflows;
- POS order ledger and configurable payment integrations;
- perpetual stock, purchasing, delivery, stocktake, waste, transfer, variance, and loss control;
- supplier price history, recipe cost, menu margin, allergens, and traceability;
- food safety, cellar, licensing, incidents, cash control, maintenance, and emergency workflows;
- utilities, energy cost, water, carbon, and operational waste;
- dashboards, exports, notifications, audit trail, backup, and restore testing.

P1 should follow after P0 stabilises:

- full rota, time and attendance, leave, payroll export, and labour forecasting;
- event and function management;
- bookings, guest CRM, feedback, review workflows, and marketing measurement;
- advanced fraud and anomaly detection;
- invoice OCR, supplier comparison, and suggested purchase orders;
- accommodation, gaming, entertainment, outdoor areas, and other conditional modules;
- knowledge graph, research capture, context packs, and governed AI assistance.

P2 and future scope:

- direct payroll submission;
- direct accounting ledger and Making Tax Digital submission unless implemented through an approved integration;
- card processing developed from first principles;
- autonomous legal decisions;
- biometric identification stored by EOS;
- facial recognition or emotion analysis;
- an AI action that changes price, stock, payment, personnel, licensing, or safety status without an authorised human approval.

5 Operating assumptions

- The first legal ruleset is England. Wales, Scotland, and Northern Ireland MUST be separate versioned rulesets because licensing, food rating display, and other duties differ.
- Each customer organisation may operate one or more premises.
- A premises can lose internet access for an entire shift.
- Devices may be shared, but each action must be attributable to one authenticated user.
- Browser storage can be cleared or exhausted. The user interface must detect risk and provide a controlled warning before data is endangered.
- Online card payments are provided by a PCI-compliant payment provider. EOS must never store full card numbers, CVV values, PINs, magnetic-stripe data, or unencrypted payment credentials.
- Legal content changes. No legal threshold or duty may be hard-coded without a source, jurisdiction, effective date, review date, and version.
- Premises licence conditions are individual. The actual licence and its conditions must be entered and approved during onboarding.
- Dates use ISO 8601 internally. Money is stored as integer minor units. Quantities use decimal values with explicit units. Timestamps are stored in UTC and displayed in the premises time zone.

6 Users and authority

6.1 Core roles

Role	Main responsibility	Default authority
Organisation Owner	Subscription, legal entity, organisations, data ownership	Full organisation administration

Role	Main responsibility	Default authority
System Administrator	Configuration, users, devices, integrations	Administration without automatic operational competence
General Manager	Premises control, compliance, P and L, approvals	Full premises operations
Duty Manager	Shift control, exceptions, cash and safety	Shift-scoped management
Cellar Manager	Cellar assets, deliveries, conditioning and dispense	Manage and validate cellar work
Kitchen Manager	Food safety, recipes, allergens, deliveries and production	Manage and validate kitchen work
FOH Staff	Service, cleaning, checks and permitted POS actions	Execute and log assigned work
BOH Staff	Preparation, temperatures, cleaning and waste	Execute and log assigned work
Stock Controller	Purchasing, counts, adjustments and investigations	Inventory control with approval thresholds
Finance User	Reconciliation, reporting, exports and budgets	Financial access without safety overrides
Maintenance User	Assets, faults, inspections and work orders	Scoped asset access
Auditor	Read-only evidence and audit review	Time-limited read access
Contractor	Assigned work order and evidence only	Least-privilege temporary access

6.2 Authorisation formula

Every protected action MUST evaluate:

allowed = active identity AND active membership AND permitted role AND current competency AND premises scope AND shift authorisation AND device trust AND rule conditions

Role membership alone must never unlock a controlled task. The system must show why access is denied and who can resolve it without revealing private information.

6.3 Permission levels

- Level 0 View
- Level 1 Execute and Log
- Level 2 Manage and Validate
- Level 3 Full Premises Access
- Level 4 Organisation Administration

Permissions MUST be granular capabilities, not only numeric levels. Examples include `stock.adjust`, `cash.refund.approve`, `allergen.publish`, `shift.open`, `licence.override`, `user.invite`, and `evidence.export`.

7 Information architecture

The mobile navigation MUST contain a maximum of five primary items:

1. Run
2. Tasks
3. Sell
4. Stock
5. More

The More area contains Food, Cellar, People, Compliance, Cash, Assets, Utilities, Waste, Events, Guests, Finance, Knowledge, Reports, and Settings according to permission.

Desktop navigation MAY use a persistent left rail. The same route and permission model must serve mobile, tablet, and desktop.

Minimum route map:

/auth
/onboarding
/run
/tasks
/sop/:sopId/run/:executionId
/sell
/orders/:orderId
/stock

/stock/items/:itemId
/stock/counts/:countId
/purchasing
/suppliers/:supplierId
/menu
/recipes/:recipeId
/food-safety
/cellar
/people
/compliance
/cash
/assets
/utilities
/waste
/events
/incidents
/finance
/reports
/knowledge
/settings
/sync

Every record screen MUST show its status, premises, owner, last update, sync state, audit access, and permitted actions.

8 First run onboarding

Onboarding is a resumable wizard. It MUST autosave locally after every field change and synchronise each completed section when online. It MUST work offline after the organisation owner has completed the initial online account and device enrolment. The wizard shows progress, missing required data, validation errors, responsible approver, and readiness impact.

The user may import CSV, XLSX, or supported integration data. Imported rows enter a review queue. No import may silently replace an approved record.

8.1 Step 1 Owner account

Required inputs:

- owner full name;
- email address;
- verified mobile number or recovery method;
- passkey registration, or password plus MFA;
- acceptance of terms and privacy information;
- recovery codes confirmation;

- default language, time zone, date format, and accessibility preferences.

The first account and tenant creation require internet connectivity. If the device is offline, show a clear explanation and allow the user to explore a non-persistent demonstration only.

8.2 Step 2 Organisation

Capture:

- legal name, trading name, company number when applicable, legal structure, UTR reference field, VAT status and VAT number;
- registered address, correspondence address, billing contact, accountant contact, insurer contact;
- organisation logo and basic display theme;
- financial year start, reporting week start, base currency, tax jurisdiction;
- data controller contact, privacy contact, retention owner;
- number of premises and total full-time-equivalent employees;
- subscription owner and emergency recovery administrators.

Sensitive identifiers must be masked by default and permission-restricted.

8.3 Step 3 Premises identity

Capture for each premises:

- trading name, full address, postcode, geolocation confirmation, telephone, public email, website;
- country and legal jurisdiction, local authority, police licensing area, premises time zone;
- rateable value when used for licence administration;
- ownership or tenancy, landlord or pub company, lease obligations, insurer policy references;
- normal opening hours, alcohol sale hours, late-night refreshment hours, seasonal variations;
- maximum capacity, fire assembly points, public entrances, staff entrances, accessible routes;
- public and non-public zones, rooms, floors, gardens, kitchen, cellar, stores, office, plant areas, toilets, car park, delivery points, smoking area, and accommodation;
- table, bar, till, stock, meter, evacuation, and cleaning zone identifiers.

Upload a premises plan and allow zones and asset locations to be mapped to it.

8.4 Step 4 Business activities

Use yes, no, and not yet values for:

- alcohol on-sales and off-sales;

- food preparation, hot food after 23:00, takeaway and delivery;
- live music, recorded music, films, dancing, indoor sport, boxing or wrestling;
- gaming machines, pool, darts, skittles, televised sport;
- functions, weddings, ticketed events, outside service, pavement use;
- accommodation, staff accommodation, play areas, EV charging;
- home delivery, online ordering, customer accounts, gift cards;
- brewing, cider making, wholesale alcohol sales, or transfers between legal entities.

Answers activate conditional modules and compliance applicability tests.

8.5 Step 5 Licensing

Capture and require manager approval of:

- premises licence number, issuing authority, issue date, annual fee date, licence summary, full licence, plan, and operating schedule;
- DPS name, personal licence number, issuing authority, appointment date, contact method, and evidence;
- all licensable activities, permitted days and hours, seasonal variations, non-standard timings, and capacity conditions;
- every premises-specific condition as a structured rule with source page, exact wording attachment, plain-language explanation, responsible role, evidence, frequency, and escalation;
- age verification policy, accepted identification, refusal process, incident process, noise controls, door supervision, CCTV conditions, dispersal policy, and children restrictions;
- temporary event notices and time-limited permissions;
- TheMusicLicence or other applicable permissions;
- gaming machine notifications or permits when applicable.

The user must sign a declaration that the structured conditions match the current licence. A second authorised reviewer SHOULD approve them. The system must never infer missing licence conditions from another pub.

8.6 Step 6 Food business and allergens

Capture:

- food business registration authority, registration date, evidence, latest inspection, rating and next review;
- food safety management system selection and documents, including SFBB or an equivalent HACCP system;
- food activities and production methods;
- responsible food safety lead and deputy;
- refrigeration, freezing, cooking, reheating, hot holding, cooling, delivery, and probe calibration controls;

- cleaning method, chemicals, contact times, PPE, safety data sheets, and storage;
- pest control contractor, visits and emergency contact;
- waste contractor and food waste arrangements;
- allergen policy, the 14 regulated allergens, cross-contamination controls, PPDS use, label approval, recipe change approval, and emergency response;
- suppliers, traceability contacts, recall contact, quarantine area, and withdrawal process;
- water source and any private water supply controls.

The food operation cannot be marked ready until critical controls, equipment, responsible people, and daily diary tasks are configured.

8.7 Step 7 Stock and suppliers

Capture or import:

- suppliers, contacts, account numbers, order days, delivery days, lead times, minimum orders, delivery charges, payment terms, returns policy, and approved status;
- alcohol wholesaler Unique Reference Number and dated AWRS check evidence where applicable;
- supplier SKUs, pack sizes, units, conversion factors, barcodes, tax treatment, deposits, discounts, rebates, and price history;
- storage locations, count sheets, categories, par levels, reorder levels, maximum levels, shelf life, batch tracking, and temperature requirements;
- opening physical quantities and valuation method;
- approved substitute products and allergen restrictions.

Opening stock requires a signed stocktake. Imported opening balances must not create fictitious purchase history.

8.8 Step 8 Menu recipes and selling

Capture or import:

- sales categories, service locations, tables, revenue centres, till devices, printers and kitchen display destinations;
- products, sizes, measures, modifiers, inclusive tags, prices, VAT classification, service charge, tips treatment, and availability;
- recipes and sub-recipes with ingredients, gross quantity, yield, waste factor, portion quantity, preparation steps, equipment, allergens, provenance, and effective dates;
- draught products with container size, expected saleable yield, line-clean allowance, head or serve standard, and legal measure configuration;
- price bands for eat-in, takeaway, event, happy hour, staff, and customer accounts;
- payment provider, accounting integration, cash drawers, floats, refund limits, offline tender policy, and receipt details.

Publishing a menu requires an allergen review and margin review. A changed ingredient must identify every affected recipe and item before publication.

8.9 Step 9 People and competency

Capture:

- employee identity, preferred name, contact details, emergency contact, start date, employment type, contracted hours and role;
- right-to-work check type, check date, checker, evidence access restriction, expiry and follow-up date;
- age and young-worker restrictions where relevant without displaying date of birth unnecessarily;
- skills, certificates, training, assessments, expiry, refresher rules, trainer, and attached evidence;
- personal licence status, first aid, fire marshal, food safety, allergens, manual handling, COSHH, cellar, equipment, safeguarding, conflict management, and other configured competencies;
- till permissions, refund limits, stock adjustment limits, keyholder status, shift roles, and premises scope;
- availability, rota preferences, payroll identifier, and consent or privacy acknowledgements.

Staff can be invited by email or enrolled on-site by an authorised manager. No user receives a safety-critical permission until the required competency is active.

8.10 Step 10 Property assets and safety

Capture:

- asset register with category, manufacturer, model, serial number, location, owner, installation date, warranty, service interval, inspection requirement, contractor, and documents;
- fire risk assessment, responsible person, alarm, detection, extinguishers, emergency lighting, evacuation plan, drills, and service dates;
- gas equipment, interlocks, ventilation, service records and emergency isolation;
- electrical installation, portable equipment policy, emergency isolation and inspections;
- asbestos register and management plan when applicable;
- Legionella assessment, outlets, flushing, temperatures and actions;
- first-aid assessment, kits, AED, trained people and inspection schedule;
- COSHH inventory, safety data sheets, risk assessments, storage and spill controls;
- security, keys, access control, panic buttons, CCTV, alarms, lone working and violence controls;

- lift, hoist, cellar drop, pressure system, refrigeration, extraction, play equipment and other applicable controls.

Each applicable asset creates its maintenance and evidence schedule.

8.11 Step 11 Utilities carbon and waste

Capture:

- electricity, gas, water, LPG, oil, biomass, refrigerants and other sources;
- meter and submeter hierarchy, serial number, unit, multiplier, location, supplier, tariff, standing charge, contract dates and reading method;
- opening readings with photograph and date;
- approved greenhouse gas conversion factor set, year, geography and source;
- waste streams, bin locations, collector, carrier registration evidence, frequency, costs, transfer-note method and destination;
- business FTE count and Simpler Recycling applicability date;
- energy and water targets and normalisation measures such as opening hours, covers, revenue and service type.

8.12 Step 12 Procedures and schedules

Activate templates for opening, closing, shift checks, toilets, food safety, cellar, line cleaning, delivery, stocktake, cash, maintenance, fire, incident, waste, and utilities.

For every procedure, confirm:

- version, owner, purpose and premises scope;
- trigger or recurrence;
- assigned role and competency;
- ordered steps and dependencies;
- acceptable values and evidence;
- failure workflow and escalation;
- sign-off and second-person verification;
- review date and training impact.

8.13 Step 13 Notifications and escalation

Configure duty phone, manager contacts, quiet hours, emergency contacts, escalation ladder, notification channels, severity rules, acknowledgement deadlines, and fallback contacts.

Critical safety alerts must be visible in-app and may also use push, SMS, or email. A notification is not complete until acknowledged or escalated.

8.14 Step 14 Data protection and retention

Configure:

- data controller and processor information;
- purposes, lawful bases, retention periods and restricted categories;
- CCTV purpose, signage, access, retention, disclosure and subject-access process;
- incident evidence permissions;
- personnel file permissions;
- audit export permissions;
- device loss process, breach response contacts, backup region and deletion policy.

8.15 Step 15 Review and launch

Show a readiness score by domain, but never use one score to hide a critical failure. Statuses are:

- Not configured
- In progress
- Ready with advisory actions
- Blocked by critical action
- Approved for operation

The launch screen MUST list every blocker, owner, due date, and direct action. It must require owner approval and premises manager approval. The system then creates the first opening tasks and a supervised demonstration shift.

9 Authentication and account security

9.1 Online authentication

The production implementation MUST support:

- passkeys using WebAuthn as the preferred method;
- email and password as a fallback with breached-password screening and a minimum 12-character policy;
- TOTP or authenticator-based MFA for owners, administrators, managers, finance users, and users with sensitive exports;
- optional platform biometric unlock through WebAuthn without storing biometric data;
- verified email and recovery method;
- rotating refresh tokens, short-lived access tokens, session revocation, device list, and remote sign-out;
- rate limits, bot protection, suspicious-login alerts, and generic error messages that do not disclose whether an account exists;

- single use invitations with expiry, intended email, premises scope, role proposal, and manager approval;
- recovery codes and a controlled account recovery workflow;
- an emergency owner recovery process that cannot be completed by one support operator alone.

Password hashes MUST use Argon2id with current reviewed parameters. Secrets MUST never be logged. Authentication events enter the audit trail.

9.2 Shared devices

A premises tablet may be enrolled as a trusted shared device. Each user still signs in or unlocks individually. Fast user switching may use a staff code plus a device-bound credential, but a simple staff code alone must not authorise manager, finance, export, or user-administration actions.

Shared devices MUST:

- lock after a configurable inactivity period, default two minutes for operational screens and immediately after a manager action if handed back;
- conceal previous users' restricted notifications and drafts;
- identify the current user prominently;
- require step-up authentication for refunds, void approvals, stock adjustments, cash declarations, licence overrides, sensitive incidents, exports, and configuration;
- prevent browser autofill from exposing another user's credentials.

9.3 Offline unlock

Offline authentication is an unlock of a previously verified local profile, not a new server login.

After successful online authentication and device enrolment, the app stores a minimal encrypted offline profile containing user ID, display name, premises scope, last verified role and competency claims, offline expiry, and a device-wrapped key. The user may unlock using a platform authenticator or a device-specific PIN. The PIN is processed through a slow key derivation function and is never stored or sent to the server.

Default offline access validity is seven days from the last verified online authentication.

Administrators may shorten it by risk class. When it expires, the app becomes read-only except for a 12-hour emergency continuity session activated by an already-enrolled duty manager with a reason and, where another authorised person is available, a second sign-off.

Offline mode MUST NOT allow:

- creating an organisation or user;
- accepting an invitation for the first time;
- resetting credentials or MFA;

- changing roles, permissions, competencies, licence rules, retention rules, or approval thresholds;
- exporting restricted data;
- changing bank, payment, payroll, or accounting integrations;
- approving a queued action that the user's last verified claims did not allow.

If a user was revoked while a device was offline, the next sync locks that local profile. Existing offline records are preserved and marked for manager review. They must not be deleted because doing so would destroy operational evidence.

9.4 Session and step up rules

- Access tokens expire in 10 minutes or less.
- Refresh tokens rotate and are bound to a device session.
- Manager and finance actions require authentication freshness of 15 minutes.
- Viewing highly sensitive incidents or right-to-work evidence requires step-up authentication.
- Three failed offline PIN attempts introduce progressive delay. Ten failures lock offline access until online recovery by an authorised manager.
- Device time tampering beyond tolerance must be flagged. Server time remains authoritative after sync.

10 Offline first architecture

10.1 Client storage

Use a local encrypted SQLite-compatible database in WebAssembly with Origin Private File System support where available. Provide an IndexedDB persistence adapter where OPFS is unavailable. The application must use one repository interface so business logic does not depend on the storage adapter.

The local database contains only the premises data needed by the enrolled user and device. Encrypt sensitive field values using Web Crypto with non-extractable keys where supported. Cache the application shell, current procedures, active tasks, permitted product catalogue, required configuration, and recent operational records.

The service worker **MUST** pre-cache the application shell and a versioned critical-offline bundle. It must never cache authentication responses, restricted exports, full personnel files, or raw payment data.

10.2 Local transaction pattern

Every local mutation **MUST** occur in one atomic transaction that:

1. validates the command against local rules and permissions;
2. writes the domain record or append-only event;
3. writes an audit event;
4. writes an outbox item with a UUIDv7 idempotency key;
5. updates the local read model;
6. returns a visible Local only status until acknowledged by the server.

The app must never show Synced until the server has accepted the operation and returned its server version.

10.3 Synchronisation triggers

Attempt synchronisation:

- after login and offline unlock when connectivity exists;
- when the browser reports online;
- when the app returns to the foreground;
- every 30 seconds while active and changes remain queued;
- through Background Sync where supported;
- when the user presses Sync now;
- before logout, shift handover, close, export, and software update.

Background Sync is an enhancement, not the only mechanism. The application must work on browsers that do not support it.

10.4 Outbox and retries

Outbox states are Pending, Uploading, Acknowledged, Conflict, Rejected, and Quarantined.

- Retry transient failures with exponential backoff and jitter.
- Do not retry validation, permission, or schema failures automatically.
- Keep idempotency keys for at least the server audit retention period required to prevent duplicate financial or stock effects.
- Sync events in causal order within an aggregate.
- Independent aggregates may sync in parallel with a configurable limit.
- A failed item must not block unrelated items unless they share a dependency.
- The Sync Centre must show counts, age of oldest item, last success, storage usage, error explanation, and safe recovery action.

10.5 Attachments and deferred upload

Photos, scans, signatures, audio notes, and permitted video are stored locally as encrypted blobs with a local manifest and content hash. Capture must succeed without connectivity.

When internet becomes available:

1. upload metadata and obtain an upload session;
2. upload the blob in resumable chunks;
3. verify size and SHA-256 hash;
4. attach the server object reference to the domain record;
5. mark the local item acknowledged;
6. retain or purge the local blob according to policy only after verification.

Critical evidence photographs upload over any allowed connection by default. Large video waits for Wi-Fi unless the user or manager permits mobile data. The user must see queued size and may pause non-critical uploads. App uninstall or storage clearance warnings must explain that Local only records could be lost.

10.6 Conflict policy

Data type	Conflict policy
Sales, payments, stock movements, readings, checklist answers, incidents, audit events	Append only and idempotent; never overwrite
Count line for the same item and count session	Preserve both submissions and require reconciliation
Approval or sign-off	Immutable event; a later reversal is a new event
Procedure execution	Merge distinct completed steps; conflicting answer to one step requires review
Draft note or description	Three-way merge when possible; otherwise user chooses
Master data such as price, recipe, rule or role	Versioned optimistic concurrency; rejected stale edits require review
Attachment	Content-addressed; duplicate hashes link once, different files both survive
Deletion	Tombstone with version and retention; never physical deletion during ordinary sync

Last-write-wins MUST NOT be used for money, stock, safety evidence, competency, permissions, licence conditions, or sign-off.

10.7 Schema and software upgrades

Local migrations must be forward-only, transactional, tested with existing queued data, and recoverable from a pre-migration encrypted backup. Do not activate a new client version if it cannot read or safely migrate the current outbox. Forced upgrades must be postponed during an active shift unless a critical security issue requires action and a continuity route is provided.

10.8 Offline payment policy

Cash and approved account tenders may be recorded offline. Card store-and-forward may be offered only through a payment provider that explicitly supports it, under that provider's risk limits. EOS must show the payment as Pending authorisation until the provider confirms it. EOS must never simulate approval or describe an unverified offline card transaction as paid.

11 Technical architecture

11.1 Reference stack

Antigravity may use equivalent technologies if it records an ADR and preserves all behaviours. The preferred stack is:

- TypeScript throughout;
- React with Next.js or an equivalent PWA-capable framework;
- a service worker built with Workbox or equivalent;
- SQLite WASM plus OPFS with IndexedDB fallback on the client;
- PostgreSQL on the server;
- an API layer using REST with OpenAPI 3.1, or typed RPC plus generated API documentation;
- object storage with short-lived signed upload sessions and malware scanning;
- a durable queue for notifications, integration events, file processing, and reports;
- OpenTelemetry for traces, metrics, and logs;
- infrastructure defined as code;
- automated migrations and rollback-safe deployments.

11.2 Service boundaries

Begin as a modular monolith with strict modules rather than premature microservices:

- Identity and Access
- Organisation and Premises
- Rules and Compliance
- Tasks and SOPs
- Evidence and Incidents
- People and Competency

- POS and Cash
- Inventory and Purchasing
- Menu and Recipes
- Food Safety
- Cellar
- Assets and Maintenance
- Utilities Carbon and Waste
- Finance and Reporting
- Notifications and Integrations
- Knowledge and AI
- Synchronisation

Modules communicate through typed commands, queries, and domain events. A module may not write another module's tables directly.

11.3 Multi tenancy

Every tenant-owned record MUST contain `organisation_id`; premises records also contain `premises_id` where relevant. Enforce tenant isolation in application code and PostgreSQL row-level security. Background jobs and exports must carry an explicit tenant context. Tests must attempt cross-tenant reads and writes.

11.4 Event and audit model

Important domain actions publish immutable events. Audit records contain:

```

event_id: uuidv7
organisation_id: uuid
premises_id: uuid_or_null
aggregate_type: string
aggregate_id: uuid
aggregate_version: integer
action: string
actor_user_id: uuid_or_system
actor_device_id: uuid_or_null
occurred_at_client: timestamp
received_at_server: timestamp
previous_state_hash: string_or_null
new_state_hash: string_or_null
reason_code: string_or_null
reason_text: string_or_null
source: online_offline_import_integration_system
idempotency_key: uuid
correlation_id: uuid

```

classification: internal_confidential_restricted

Audit events must be tamper-evident through chained hashes or an equivalent reviewed design.
Corrections create new events.

12 Universal control engines

12.1 Rules and compliance engine

A rule contains:

- stable ID, title, domain, jurisdiction and premises applicability;
- classification as legal, licence, insurer, landlord, company, manufacturer, or best practice;
- authoritative source URL or uploaded source, citation, exact source date, effective date, expiry, last review and next review;
- applicability expression;
- task trigger, recurrence, deadline, grace period and permitted postponement;
- responsible role, required competency and approver;
- required evidence and retention;
- pass conditions, failure severity, corrective action and escalation;
- version and change notes.

Rule updates never rewrite historic tasks. New versions apply from their effective date and create an impact assessment for procedures, training, forms, menus, schedules, and reports.

12.2 Task engine

Tasks may be scheduled, event-triggered, manually created, or produced by a failed control. Each task contains owner, due window, priority, dependency, location, procedure version, evidence requirement, status and escalation. Task statuses are Planned, Available, In Progress, Blocked, Failed, Awaiting Review, Complete, Missed, Cancelled, and Superseded.

The engine must handle recurrence in the premises time zone, daylight-saving changes, missed instances, shift reassignment, and explicit cancellation reason.

12.3 Executable SOP engine

Each SOP version contains purpose, scope, trigger, hazards, competency, PPE, tools, estimated time, preconditions, ordered steps, decision gates, expected results, failure states, corrective actions, escalation, evidence, sign-off, metrics, change summary and review date.

Step types include:

- instruction acknowledgement;
- checkbox;

- number with unit and tolerance;
- temperature;
- date and time;
- text or reason;
- single or multiple choice;
- barcode or QR scan;
- photograph or document;
- signature;
- timer;
- dependent step;
- two-person verification;
- equipment reading;
- stock movement;
- pass or fail assessment.

Conditional logic must be deterministic, versioned and testable. A failed safety gate must not be bypassed by a generic Complete button.

12.4 Evidence engine

Evidence links to the rule, task, SOP version, step, actor, device, timestamp, premises, location, and source. Evidence stores a content hash and chain of custody. Restricted evidence uses separate access policy and retention. Redaction creates a derivative while preserving the restricted original.

12.5 Exception and corrective action engine

A failed value creates an exception with severity, immediate containment, affected products or assets, responsible person, due time, root cause, corrective action, preventive action, verification and closure approval. Critical exceptions appear on Run the Pub and may block opening, service, a menu item, an asset, or the whole premises.

12.6 Competency engine

Competency records contain requirement, person, trainer, learning content, assessment, result, evidence, issue date, expiry, refresher, supervised status, premises scope and permitted tasks. Procedure changes generate a training impact list. Expired competency removes access at the effective time and reassigns affected tasks.

12.7 Notification engine

Notifications are events with severity, audience, channel, acknowledgement requirement, deadline, escalation and deduplication key. Acknowledgement and resolution are different states. Critical notifications escalate until a qualified person acknowledges them.

13 Run the Pub and shift control

13.1 Control board

Run the Pub is the default screen during operations. It shows:

- premises state Closed, Opening, Open, Restricted, Emergency, Closing, or Locked;
- current duty manager and key competent staff;
- critical blockers and overdue tasks;
- food safety, licensing, fire, staffing, stock, cash, assets, utilities, bookings and incidents;
- current sales, labour, waste, variance signals and service alerts;
- connectivity, last successful sync, queued items and local storage risk;
- handover notes and next scheduled controls.

Readiness is shown by domain. A red critical domain cannot be averaged into a green overall score.

13.2 Opening

Opening creates role-specific FOH, BOH, cellar, cash, safety, external sweep, toilet, equipment and temperature tasks. The system blocks Open when any configured critical gate has failed or lacks evidence. An override is available only where the governing rule permits it and requires authority, reason, containment, expiry and audit.

13.3 Live shift

The control board updates from tasks and operational events. It must support staff assignment, service mode, 86 status, equipment failure, delivery arrival, emergency mode, task escalation, and an operational timeline.

13.4 Handover

Handover includes staffing, cash, stock shortages, equipment faults, temperatures, bookings, incidents, restricted guests, accessibility arrangements, deliveries, maintenance, compliance tasks, keys and unresolved exceptions. Both outgoing and incoming managers sign where practicable.

13.5 Closing

Closing enforces configured sequence, including grill scraping before floor mopping, cash reconciliation, food disposition, refrigeration checks, stock security, gas and electrical checks, doors, alarm, key transfer and final premises state. Unfinished critical items block closure or create a documented emergency closure route.

14 POS orders and payments

The POS MUST support product search, category tiles, service location, table, covers, server, courses, notes, modifiers, holds, sends, transfers, merges, splits, discounts, comps, voids, refunds, receipts, customer accounts, gift cards through an integration, cash and configured payment tenders.

Each sale line stores product and recipe version, price, tax, discount, modifier, quantity, unit, server, location and stock effect. Repricing a product never alters historic sales.

Controlled behaviours:

- alcohol sale is blocked unless the user has current shift authorisation and required training;
- products and measures must follow the approved premises configuration;
- unavailable or quarantined ingredients automatically affect menu availability according to recipe substitution rules;
- allergen data is available before the order is confirmed, but staff must not promise absence of cross-contamination unless the approved policy supports it;
- refunds, discounts, comps and voids above thresholds require reason and approval;
- no order may be silently deleted;
- offline card status is explicit;
- end-of-day totals reconcile order, tender, provider, drawer and bank-deposit records.

15 Inventory purchasing and variance

15.1 Inventory ledger

Inventory is event-sourced or represented by an equivalent immutable movement ledger. Movement types include delivery, sale consumption, production, transfer, return, waste, comp, staff use, line clean, ullage, tasting, breakage, stocktake adjustment, quarantine, release and correction.

Each movement stores item, batch where required, quantity, unit, base-unit conversion, cost layer, location, actor, reason, source transaction and timestamp.

15.2 Theoretical stock

For each item and location:

```
theoretical closing = opening + deliveries + transfers in + production  
- theoretical sales use - recorded waste - comps - staff use - returns  
- transfers out
```

At stocktake:

$$\text{unexplained loss quantity} = \text{theoretical closing} - \text{actual closing}$$

$$\text{variance percent} = \text{unexplained loss quantity} / \text{theoretical consumption} \times 100$$

When theoretical consumption is zero, show variance percentage as Not applicable and retain quantity and value variance.

Also calculate cost value, estimated retail value and gross profit lost. Show positive and negative variance separately; do not hide gains by netting them against losses.

15.3 Wet stock yield

For cask, keg, cider and spirits, configure container volume, expected saleable volume, standard measure, head allowance, line-clean loss, ullage, tasters, residual and tolerance. Compare theoretical serves with recorded serves and physical quantity. Variance analysis must identify product, day, service period, till, server and reason without automatically accusing an employee.

15.4 Stocktake

Stocktakes support blind count, count sheets, barcode scan, volume estimation, two counts, recount thresholds, frozen snapshot time, offline operation and manager approval. Lines added after the snapshot are separate events. Large differences require investigation before posting an adjustment.

15.5 Purchasing and delivery

The system supports requisition, suggested order, purchase order, approval, supplier acknowledgement, goods receipt, invoice match, credit, return and closure.

Delivery intake records supplier, PO, product, batch, use-by or best-before date, quantity, temperature where applicable, condition, leakage, label, shelf life, substitution, price and evidence. Failed intake creates quarantine or rejection. Cask and keg minimum shelf-life thresholds are configurable, with the source reference values of 14 days for cask and 20 days for keg stored as premises assumptions until approved.

15.6 Supplier price control

Store quoted, ordered, delivered and invoiced prices separately. Calculate true landed unit cost after pack size, conversion, discount, rebate and delivery charge. Every change records effective date and source. Price increases trigger affected-recipe recalculation and margin alerts.

16 Menu recipes and allergens

Recipes support sub-recipes, yield, preparation loss, portion size, effective dating, alternatives, provenance, instructions, equipment, allergens and images. Cost uses the appropriate approved supplier or weighted inventory cost according to the configured accounting method.

Calculate:

$$\text{net sales} = \text{gross sales} - \text{VAT} - \text{refunds} - \text{discounts}$$

$$\text{recipe cost} = \text{sum ingredient base quantity} \times \text{current unit cost adjusted for yield}$$

$$\text{gross profit} = \text{net selling price} - \text{recipe cost}$$

$$\text{gross profit percent} = \text{gross profit} / \text{net selling price} \times 100$$

Flag food cost above the configured threshold, default 30 percent, and display the cost driver. The historic 70 to 78 percent GP range is a configurable target, not a universal rule.

The allergen matrix is derived from approved ingredient specifications and approved substitutions. A supplier or ingredient change places affected recipes into Review Required. A menu item cannot be published if allergen review is incomplete. PPDS labels must use a versioned ingredient declaration and retain a print record.

17 Food safety

The food-safety module MUST support:

- food business registration and inspection evidence;
- HACCP or SFBB controls and daily diary;
- supplier approval and one-step-back, one-step-forward traceability where applicable;
- delivery, chilled, frozen, hot-hold, cook, reheat and cooling readings;
- configurable thresholds with approved source and corrective action;
- probe identification, calibration and failure response;
- cleaning schedules, two-stage cleaning, chemicals, dilution, contact time and PPE;
- cross-contamination and personal hygiene controls;
- opening and closing food disposition;
- date coding, day dots, FIFO, Use First and expiry;
- pest observations, visits and action;
- illness reporting and return-to-work controls;
- allergen queries, allergen incidents and emergency response;
- quarantine, withdrawal, recall, customer notification and authority reporting workflow;
- PPDS label approval and history.

A failed refrigeration reading must identify affected stock and require disposition. Completing the reading does not complete the control.

18 Cellar and dispense

The cellar module includes deliveries, cask and keg inventory, stillaging, venting, conditioning, CATT, connection, line allocation, line cleaning, cellar temperature, ventilation, glassware, pour checks, yields, ullage and waste.

Cask workflow:

1. record delivery and validate label, leakage, shelf life and condition;
2. record rolling and stillaging with level or chocking evidence;
3. schedule venting 2 to 4 hours later according to the approved product procedure;
4. start the approved conditioning timer, with 72 hours as a configurable reference value;
5. record Clarity, Aroma, Taste and Temperature;
6. unlock Connect to Line only when the timer is complete and CATT passes;
7. record line, date, expected yield and person connecting.

Line cleaning requires line selection, product disposition, chemical, concentration, PPE confirmation, start and contact timer, rinse confirmation, test where required, reconnection, volume loss and sign-off. PPE photographs must be supported where the premises procedure requires them.

The system schedules daily cellar ventilation and the configured 10-minute manual ventilation task where adopted. Reference beverage head and pour specifications remain premises-configurable.

19 Licensing safeguarding and incidents

19.1 Alcohol controls

Support digital DPS authorisation per person or approved shift arrangement. POS alcohol service remains blocked until required authorisation is active.

Challenge 25 and refusal records include date and time, location, staff member, apparent age range, ID type, result, reason, description with data minimisation, aggression or safeguarding indicator, manager notification and follow-up. Do not require the person's name unless a lawful and necessary reason exists.

19.2 Incidents

Incident types include accident, near miss, violence, theft, fraud, fire, food safety, allergen, suspected food poisoning, spiking, safeguarding, data breach, licensing, ASB, equipment, utility failure and other.

Incidents support immediate actions, people involved, witnesses, injuries, first aid, emergency services, evidence, CCTV reference, notifications, RIDDOR assessment, insurer notification, root cause, corrective action, restricted access and closure approval.

19.3 Safeguarding and emergency guides

Provide guided modes for Ask for Angela, suspected spiking, violence, missing or vulnerable person, silent evacuation, fire, gas smell, power loss, water loss, flood, refrigeration failure, cyber incident and severe staff shortage. Guidance content is versioned and approved. The source CHAHREH material must be retained as a reference issue until its exact authoritative source and wording are reviewed.

20 People rota and workforce controls

The people module supports profiles, contracts metadata, roles, competencies, right-to-work controls, rota, availability, clock-in and out, breaks, leave, sickness, overtime, young-worker restrictions, labour budgets, payroll export, tips and service-charge records, performance processes and leaver workflow.

Restricted personnel records are separated from operational profiles. A line manager may see operational competence without seeing immigration, health, disciplinary, or payroll evidence unless specifically authorised.

The leaver workflow revokes sessions, disables offline profiles at next sync, collects keys and devices, reassigns tasks, preserves audit history and schedules retention or deletion.

21 Cash fraud and reconciliation

Support tills, drawers, floats, safe drops, paid-outs, cash counts, card batches, provider settlements, bank deposits, refunds, voids, discounts, comps, gift cards, customer accounts and chargebacks.

Cash declarations may use blind entry. Variances require reason codes and thresholds. Dual control is required above configured amounts. The system identifies exception patterns such as unusual void rate, discount concentration, repeated drawer shortage, after-hours activity and stock loss correlation. These are investigation signals, not automatic findings of misconduct.

22 Assets maintenance health and safety

Assets have status Active, Restricted, Out of Service, Quarantined, Disposed, and Replaced. Work orders contain fault, risk, containment, priority, contractor, parts, cost, downtime, evidence, verification and return-to-service approval.

The module creates schedules for fire, gas, electrical, extraction, refrigeration, water hygiene, asbestos, lifting equipment, pressure systems, security, play equipment and other applicable assets. It also supports risk assessments, COSHH, safety data sheets, first aid, accidents, RIDDOR assessment, manual handling, lone working, slips and trips, work at height, contractor control and permits to work.

An asset with a critical safety failure must be unavailable in the related procedure or menu workflow until an authorised return-to-service event.

23 Utilities carbon and waste

23.1 Metering and cost

Store manual, imported and smart readings with actual or estimated status. Validate monotonic readings while allowing documented meter replacement, rollover and correction. Calculate consumption using the meter multiplier, tariff period, unit rate, standing charge and taxes where configured.

Report electricity kWh, gas kWh, water cubic metres, cost and intensity by opening hour, cover, £1,000 net revenue and relevant service period.

23.2 Carbon

Store conversion factor set, factor ID, activity unit, kgCO₂e per unit, scope, source organisation, publication year, effective period and source URL. Never overwrite a factor used in historic calculations. Recalculation creates a new reporting version.

Classify direct fuels and refrigerants as applicable Scope 1 sources, purchased electricity as Scope 2, and selected value-chain categories as Scope 3. The screen must state which scope and methodology it uses.

23.3 Operational waste

Track food, glass, metal, plastic, paper and card, residual, cooking oil, hazardous, electrical, batteries, vapes, garden and other streams. Store quantity, unit, cost, contamination, collector, destination, transfer note, carrier check and carbon proxy where approved.

Food waste records link to inventory and cost. The old £1.27 per kg figure may exist only as a configurable assumption with date and source, not as a fixed universal value.

23.4 Energy procedures

Include opening and closing equipment controls, kill-switch audits, extraction use, hot cupboards, hobs, cellar cooling, condenser cleaning, TRV checks, leaks, abnormal base load and tariff alerts. Projected savings must display their assumptions and must not be recorded as realised savings until meter or bill evidence confirms them.

24 Finance and profitability

The finance module is a management ledger and integration layer in P0. It supports sales summaries, VAT classifications, COGS, stock valuation, purchases, waste, variance, discounts, labour, utilities, contribution, budget and cash forecast. Formal tax submissions require a reviewed accounting integration or later compliant accounting module.

Report by product, category, revenue centre, service period, day, shift and event:

- gross and net sales;
- VAT;
- COGS;
- GP value and percentage;
- recorded waste;
- unexplained stock variance;
- discounts, comps and refunds;
- labour allocation;
- energy allocation;
- contribution margin;
- budget versus actual.

The flash P and L must reconcile to its source transactions and show incomplete or unsynchronised data. It must never present a provisional offline figure as final.

25 Events guests and feedback

Events support enquiry, provisional booking, deposit, contract, capacity, package, menu, allergens, staffing, equipment, supplier orders, risk assessment, margin gate, final payment, delivery, incident link, feedback and post-event review.

Guest records use consent and purpose controls. Feedback enters a structured theme, severity, owner, response deadline, improvement action and outcome. Review replies require approval according to role. Marketing consent is separate from operational booking communication.

26 Knowledge capture and AI

EOS Capture and the Chrome extension are P1. Capture URL, title, author, publisher, date, selected text, content, screenshot, OCR, note, provenance and content hash. Classify as New, Duplicate, Update, Related, or Contradictory. Never silently discard a capture.

AI may summarise, classify, suggest relationships, identify possible duplicates, draft reports, explain a procedure, or propose actions. AI output is labelled as a suggestion with model, prompt version, context sources and time. It cannot approve safety, legal, financial, personnel, allergen, or licence decisions.

Sensitive content enters AI context only when the user's permission and the approved context policy allow it. Provide a provider abstraction and a mode in which the core pub system runs without AI.

27 Data model

All mutable master records use UUID primary keys, organisation scope, created and updated timestamps, version, status, and soft-deletion or tombstone fields. Operational events are immutable.

27.1 Identity and premises entities

`organisations, premises, premises_zones, legal_entities, users, memberships, roles, permissions, role_permissions, user_role_assignments, devices, sessions, offline_profiles, invitations, mfa_methods, recovery_codes, emergency_contacts, data_retention_policies.`

27.2 Rules and operations entities

`rules, rule_versions, rule_sources, rule_applicability, procedures, procedure_versions, procedure_steps, procedure_dependencies, task_templates, task_instances, task_assignments, executions, step_responses, signoffs, evidence, exceptions, corrective_actions, escalations, notifications, acknowledgements, shifts, handovers, readiness_snapshots.`

27.3 People entities

`people, employment_records, competency_definitions, competencies, training_courses, training_attempts, right_to_work_checks, availability, rota_shifts, time_entries, breaks, leave_requests, payroll_exports, tips_periods, tips_allocations.`

27.4 Sales and cash entities

revenue_centres, service_locations, tables, product_catalogues, sale_products, product_prices, modifier_sets, orders, order_lines, order_line_modifiers, payments, payment_attempts, refunds, discounts, voids, tills, cash_drawers, cash_sessions, cash_counts, safe_drops, bank_deposits, provider_settlements, customer_accounts.

27.5 Inventory and purchasing entities

inventory_items, inventory_units, unit_conversions, storage_locations, inventory_batches, inventory_movements, inventory_balances, stocktake_sessions, stocktake_lines, variance_investigations, suppliers, supplier_items, supplier_prices, supplier_checks, purchase_requisitions, purchase_orders, purchase_order_lines, goods_receipts, goods_receipt_lines, supplier_invoices, supplier_invoice_lines, supplier_credits, transfers, waste_events.

27.6 Menu and food entities

ingredients, ingredient_specifications, allergens, ingredient_allergens, recipes, recipe_versions, recipe_components, menu_items, menu_versions, menu_item_recipes, menu_tags, ppds_labels, food_controls, temperature_readings, probes, probe_calibrations, cleaning_schedules, pest_events, quarantine_lots, recalls, recall_actions.

27.7 Cellar entities

cellar_containers, cellar_deliveries, cask_workflows, catt_assessments, dispense_lines, line_connections, line_clean_runs, pour_checks, yield_periods, ullage_events, cellar_readings.

27.8 Compliance incident and asset entities

licences, licence_conditions, dps_authorisations, refusal_events, incidents, incident_people, incident_evidence, cctv_references, external_reports, risk_assessments, coshh_items, safety_data_sheets, assets, asset_inspections, maintenance_plans, work_orders, contractors, permits_to_work, certificates.

27.9 Utility waste finance and event entities

utility_accounts, meters, meter_readings, tariffs, utility_bills, carbon_factor_sets, carbon_factors, carbon_calculations, waste_streams, waste_collections, carrier_checks, transfer_notes, budgets,

financial_periods, flash_pnl_snapshots, events, event_packages, bookings, deposits, guest_profiles, consents, feedback, improvement_actions.

27.10 Platform entities

audit_events, domain_events, outbox_items, sync_cursors, sync_conflicts, file_manifests, file_upload_parts, imports, import_rows, exports, integration_connections, integration_runs, webhooks, feature_flags, system_settings, knowledge_objects, relationships, capture_events, ai_runs.

27.11 Required database constraints

- Unique tenant-scoped stable codes.
- Check constraints for valid quantities, currency codes, units and states.
- Foreign keys within the same organisation and premises scope.
- Unique aggregate version and idempotency key.
- Immutable triggers or application enforcement for audit, sale, stock and sign-off events.
- Partial indexes for active tasks, unsynchronised items, expiring competencies, current prices and open exceptions.
- No floating-point columns for money.
- Effective date ranges must not overlap where only one current version is allowed.

28 API contract

All APIs use TLS, tenant context, explicit authorisation, request IDs and idempotency for mutations. Publish OpenAPI documentation and generated TypeScript clients.

Minimum endpoint groups:

POST /auth/register
POST /auth/login
POST /auth/passkey/options
POST /auth/passkey/verify
POST /auth/mfa/verify
POST /auth/refresh
POST /auth/logout
GET /me
GET /devices
DELETE /devices/:id

POST /organisations
GET /organisations/:id
POST /premises
PATCH /premises/:id

GET /premises/:id/readiness
POST /premises/:id/launch

GET /rules
POST /rules
POST /rules/:id/versions
GET /tasks
POST /tasks/:id/start
POST /executions/:id/steps/:stepId/respond
POST /executions/:id/signoff
POST /exceptions/:id/actions

POST /shifts
POST /shifts/:id/open
POST /shifts/:id/handover
POST /shifts/:id/close
GET /run-board

POST /orders
POST /orders/:id/lines
POST /orders/:id/send
POST /orders/:id/payments
POST /orders/:id/close
POST /orders/:id/refunds

GET /inventory/items
POST /inventory/movements
POST /stocktakes
POST /stocktakes/:id/counts
POST /stocktakes/:id/reconcile
GET /variance
POST /purchase-orders
POST /goods-receipts

GET /recipes
POST /recipes/:id/versions
POST /menus/:id/publish
GET /allergen-impact

POST /incidents
POST /meter-readings
POST /waste-events
GET /reports/:reportId

POST /sync/push
GET /sync/pull
POST /files/upload-sessions
PUT /files/upload-sessions/:id/parts/:part
POST /files/upload-sessions/:id/complete

28.1 Sync push

POST /sync/push accepts an ordered batch with device ID, local cursor, schema version and operations. Each operation contains idempotency key, aggregate, expected version, client time, dependencies and payload. The response reports Acknowledged, Conflict, Rejected or Retry for each operation independently and returns server versions and pull cursor.

28.2 Sync pull

GET /sync/pull?cursor= returns permission-filtered changes, tombstones, rule invalidations, user-claim expiry, and a new cursor. A cursor is opaque. The client applies the batch atomically and acknowledges only after the local transaction succeeds.

28.3 Error shape

```
{  
  "type": "https://eos.example/errors/validation",  
  "title": "Validation failed",  
  "status": 422,  
  "code": "TEMPERATURE_OUT_OF_RANGE",  
  "detail": "A corrective action is required before this control can pass.",  
  "instance": "request-id",  
  "fieldErrors": [],  
  "retryable": false  
}
```

29 Reports and exports

Minimum reports:

- shift pack and handover;
- opening and closing compliance;
- missed tasks and exceptions;
- food safety diary and temperature exceptions;
- allergen change and menu approval;
- licensing authorisations, refusals and incidents;
- stock valuation, movement, waste, theoretical versus actual, variance and yield;

- supplier price movement and purchase performance;
- recipe cost and menu margin;
- cash reconciliation and exception report;
- asset compliance and upcoming services;
- energy, water, cost, carbon and waste;
- training, competency and expiry;
- daily and weekly flash P and L;
- premises compliance evidence pack;
- audit export.

Exports support PDF for human review and CSV or XLSX for data. Each export records filters, creator, time, source versions, whether queued data was excluded, and a content hash. Sensitive exports expire and require step-up authentication.

30 Security privacy and assurance

30.1 Security requirements

- Follow OWASP ASVS and the OWASP API Security Top 10 as development baselines.
- Encrypt transport with modern TLS and server data at rest using managed keys.
- Separate secrets from code and rotate them.
- Validate all input server-side.
- Use CSP, secure cookies, CSRF protection where applicable, output encoding, dependency scanning, static analysis, secret scanning, malware scanning and signed builds.
- Apply least privilege to service accounts and storage.
- Log security events without logging secrets or excessive personal data.
- Maintain tested backups, point-in-time recovery and documented disaster recovery.
- Conduct threat modelling for offline data, shared devices, attachments, payment integrations, exports and tenant isolation.
- Complete independent penetration testing before production and after major identity or sync changes.

30.2 Privacy requirements

Maintain a record of processing, data inventory, purpose, lawful basis, access policy, retention, deletion method and processor for each personal-data category. Support subject access, rectification, restriction, export and deletion where legally applicable without destroying records that must be retained.

Minimise refusal, incident, CCTV and staff data. Use restricted partitions and fine-grained permissions. The audit log records access to highly sensitive data.

30.3 CCTV

EOS stores references and controlled extracts, not continuous CCTV recording in P0. Record camera or system, purpose, incident window, authorised extractor, chain of custody, retention, disclosure, redaction and deletion. Support signage and request workflows.

30.4 Availability and recovery

Target server availability is 99.9 percent monthly excluding scheduled maintenance. The offline client is the continuity method for ordinary connectivity loss. Server recovery point objective is 15 minutes and recovery time objective is four hours for P0, subject to final commercial review. Restore tests run at least quarterly.

31 Accessibility and usability

Meet WCAG 2.2 AA. All workflows must work by keyboard and screen reader. Provide visible focus, adequate contrast, meaningful labels, error summaries, touch targets of at least 44 by 44 CSS pixels where practical, reduced motion, text resizing, and alternatives to colour-only status.

The interface uses simple black and white surfaces, rounded cards, restrained shadows and one semantic status colour set. Safety status is expressed by icon, text and colour. Do not hide critical information in hover states.

Operational screens must be usable with gloves or wet hands where practical, avoid dense typing, support barcode and camera input, and preserve a draft when interrupted.

32 Performance and reliability

- Cached application shell becomes interactive within three seconds on a representative mid-range device.
- Local commands confirm within 300 milliseconds at the 95th percentile, excluding image processing.
- Online API reads complete within one second at the 95th percentile under normal load.
- The client supports at least 10,000 queued lightweight operations and 2 GB of queued evidence subject to browser quota and device capacity.
- The app warns at 70 percent local quota, blocks non-critical large capture at 90 percent, and never purges unacknowledged records automatically.
- Sync resumes after interruption without duplicate sales, stock movements, tasks, or files.
- A premises can complete a full eight-hour offline demonstration shift and reconcile after reconnection.

33 Observability and support

Capture service health, API latency, error rate, sync backlog, oldest queued operation, conflicts, failed uploads, notification delivery, rule generation failures, backup status and integration health. Never send restricted business content to telemetry by default.

Provide an administrator support bundle containing app version, schema version, device type, storage health, redacted error codes and sync metadata. It must exclude passwords, tokens, evidence content, incident narratives and personal records.

34 Legal compliance architecture

The software must operationalise applicable requirements but must not claim that a generic template guarantees legal compliance. Before a premises launches, an authorised operator must confirm its jurisdiction, activities, licence and local conditions. The production ruleset must receive review by an appropriately qualified UK licensing, food safety, employment, data protection and health and safety adviser.

The initial source register MUST include current official pages for:

- food business duties, registration, food safety management, allergens and traceability: <https://www.gov.uk/running-food-business>
- SFBB: <https://www.gov.uk/government/publications/safer-food-better-business-sfbb/safer-food-better-business-sfbb>
- premises licensing and DPS: <https://www.gov.uk/find-licences/premises-licence>
- AWRS due diligence: <https://www.gov.uk/guidance/the-alcohol-wholesaler-registration-scheme-awrs>
- workplace health and safety: <https://www.hse.gov.uk/simple-health-safety/>
- RIDDOR: <https://www.hse.gov.uk/riddor/>
- workplace fire safety: <https://www.gov.uk/workplace-fire-safety-your-responsibilities>
- CCTV and video surveillance: <https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/cctv-and-video-surveillance/>
- right-to-work checks: <https://www.gov.uk/government/publications/right-to-work-checks-employers-guide>
- minimum wage records: <https://www.gov.uk/national-minimum-wage/employers-and-the-minimum-wage>
- workplace recycling in England: <https://www.gov.uk/guidance/simpler-recycling-workplace-recycling-in-england>

- UK greenhouse gas conversion factors for 2026:
<https://www.gov.uk/government/publications/greenhouse-gas-reporting-conversion-factors-2026>

Each source entry has an owner and review schedule. A source change creates a legal review task, not an automatic silent rules change.

35 Seed procedures

Create complete, editable, versioned templates for at least:

1. FOH opening
2. BOH opening
3. FOH closing
4. BOH closing
5. hourly toilet and zero-debris check
6. refrigeration and freezer readings
7. food delivery intake
8. cook, reheat, hot hold and cooling controls
9. end-of-day food disposition and day dots
10. allergen query and recipe change
11. product withdrawal and recall
12. cask delivery and stillaging
13. cask venting, conditioning and CATT
14. connect to line
15. weekly line clean
16. cellar temperature, ventilation and FIFO
17. glassware renovation and reservoir change
18. perfect pour check
19. Challenge 25 and refusal
20. DPS shift authorisation
21. incident and CCTV extraction
22. suspected spiking response
23. fire evacuation
24. gas smell
25. power, water and refrigeration failure
26. waste quick log and bin audit
27. stocktake and variance investigation
28. till open, safe drop and close
29. delivery receipt and invoice discrepancy
30. equipment kill-switch audit
31. asset fault and return to service
32. shift handover

- 33. event readiness and post-event review
- 34. data breach response
- 35. lone working and personal safety

Seed content must be clearly marked Template requiring premises approval. No template becomes active until reviewed.

36 Acceptance scenarios

36.1 Onboarding

- An owner creates an account online, enrolls MFA and starts a premises.
- The wizard is interrupted, reopened offline, and resumes without lost fields.
- The owner enters business activities and sees only applicable module questions.
- Missing premises licence conditions, food controls or critical assets block launch with an exact explanation.
- Imported stock and menu rows display validation and allergen-impact errors before approval.
- Two authorised users approve launch and the first operational tasks appear.

36.2 Authentication

- A known staff member unlocks an enrolled tablet with no internet and performs allowed tasks.
- A first-time user cannot accept an invitation offline.
- An expired offline profile enters read-only mode and can use the audited emergency route only as designed.
- A revoked user reconnects and is locked; their existing queued records survive for review.
- A staff PIN cannot approve a manager refund without step-up authentication.

36.3 Offline capture and upload

- Disable connectivity for eight hours during a demonstration shift.
- Record checklist answers, temperatures, photographs, cash orders, waste, stock movements, an incident and a handover.
- Close and reopen the browser and device; all local records remain.
- Restore connectivity; all operations upload once, in valid order, with attachments and hashes.
- Interrupt a large upload; it resumes from the last accepted chunk.
- Create a conflict on a stock count line; both values remain and a reconciliation task appears.

36.4 Operational gates

- An unsafe fridge reading creates an exception, identifies affected stock and blocks opening until the approved corrective route completes.
- Floor mopping is unavailable until grill scraping is recorded.
- Connect to Line stays locked until conditioning is complete and CATT passes.
- Alcohol products remain unavailable to a user without current authorisation.
- An expired allergen competency prevents publication and allergen sign-off.

36.5 Stock and profit

- A sale posts recipe consumption once.
- Delivery, waste, staff use and transfer events reconcile to theoretical stock.
- A physical count calculates unexplained loss in quantity, cost, retail value and variance percentage.
- A supplier price rise updates future recipe cost, preserves historic cost and raises a margin alert.
- Flash P and L identifies unsynchronised data and reconciles after sync.

36.6 Security and tenancy

- A user cannot access another organisation by changing an ID.
- Restricted incident and right-to-work evidence require explicit permission and step-up authentication.
- Duplicate sync calls do not duplicate money or stock.
- Deleting a record creates a tombstone and preserves the audit history.
- Storage exhaustion cannot silently remove unacknowledged evidence.

37 Test strategy

Automate:

- unit tests for formulas, thresholds, recurrence, permission and state machines;
- property-based tests for units, money, stock conservation and idempotency;
- integration tests for database constraints, files, notifications, payment and accounting adapters;
- contract tests for every API and integration;
- end-to-end tests for onboarding, opening, sale, delivery, stocktake, incident, closing and export;
- offline browser tests with network loss, reload, process termination, quota pressure and reconnection;
- sync tests with reordering, duplicate delivery, partial failure, stale versions, revoked users and clock skew;

- security tests for tenancy, broken object-level authorisation, injection, XSS, CSRF, file upload, token theft and offline database exposure;
- accessibility tests plus manual screen reader and keyboard review;
- migration tests containing old queued data;
- backup restoration and disaster recovery exercises.

No P0 release may ship with a failing critical workflow, permission, tenant-isolation, migration, offline, or sync test.

38 Delivery plan

Phase 0 Foundation

Create repository, ADRs, design system, CI, environments, database, identity, tenancy, audit, feature flags, observability and test harness.

Exit when account creation, passkey or MFA login, tenant isolation, device enrolment and an audited empty premises work.

Phase 1 Offline platform and onboarding

Build local storage adapters, encryption, service worker, offline profiles, outbox, sync API, files, Sync Centre and all onboarding steps.

Exit after the full onboarding and eight-hour offline acceptance scenarios pass.

Phase 2 Rules SOPs people and shifts

Build rules, procedures, task engine, evidence, exceptions, competency, Run the Pub, opening, handover and closing.

Exit when critical gates and role plus competency access work online and offline.

Phase 3 Stock purchasing menu and POS

Build supplier, purchase, delivery, inventory ledger, stocktake, variance, recipes, allergens, menu, orders, cash and payment adapter.

Exit when sale-to-stock-to-profit reconciliation passes without duplicate effects.

Phase 4 Compliance operations

Build food safety, cellar, licensing, incidents, assets, maintenance, utilities, carbon, waste, evidence packs and emergency modes.

Exit when the seeded operational workflows pass with reviewed England sources and premises-specific configuration.

Phase 5 People finance events and intelligence

Build rota, attendance, payroll export, advanced P and L, events, guest feedback, anomaly signals, Capture, knowledge and governed AI.

Exit when permissions, privacy boundaries and source-linked AI outputs pass review.

Phase 6 Production assurance

Complete load, accessibility, penetration, legal, data protection, backup, disaster recovery and pilot-premises testing. Run parallel operation against existing records before relying on EOS for critical control.

39 Repository deliverables

Create and maintain:

- /apps/web
- /apps/api
- /packages/domain
- /packages/ui
- /packages/offline
- /packages/sync
- /packages/schemas
- /packages/config
- /infra
- /docs/architecture
- /docs/adr
- /docs/modules
- /docs/sops
- /docs/legal-sources
- /docs/runbooks
- /tests/e2e
- /tests/offline
- /tests/security
- README.md
- CHANGELOG.md
- SECURITY.md
- PRIVACY.md
- CONTRIBUTING.md
- openapi.yaml

requirements-traceability.csv

Every module document defines purpose, screens, roles, commands, events, data, offline behaviour, permissions, notifications, audit, reports, failure states and tests.

40 Definition of done

The system is production-ready only when:

- every P0 requirement is traced to implementation and a passing test;
- onboarding configures a real premises without direct database editing;
- an enrolled team can run a complete shift offline and safely synchronise it;
- all critical actions are attributable, permission-checked and auditable;
- all attachments upload resumably with integrity verification;
- stock, purchasing, sales, waste and counts reconcile without duplicate effects;
- legal and licence rules are premises-specific, versioned, sourced and reviewed;
- critical failed controls create containment and cannot be dismissed as complete;
- reports identify provisional or unsynchronised data;
- accessibility, security, privacy, penetration, backup and restore gates pass;
- production monitoring and support runbooks are active;
- a trained manager completes pilot acceptance and signs the operational readiness record.

41 Final build directive

Start with Phase 0 and Phase 1. Generate the working application, migrations, tests and documentation, not a mock-up. Preserve the source architecture of Rules to Tasks to Evidence to Exceptions to Intelligence. Keep domain-specific pub workflows concrete. Record assumptions and ADRs. Do not claim legal compliance until the premises configuration and professional reviews required by this specification have been completed.

When a requirement is too large for one code change, split it into linked issues and continue through the phase. Do not ask for repeated permission to proceed. Stop only for a decision that changes legal scope, payment liability, data ownership, or another material product boundary that cannot safely be inferred.